

Problem

Calculate the values of R , L , and C that will result in $R = 12 \text{ k}\Omega$, $L = 40 \text{ }\mu\text{H}$, and $C = 300 \text{ nF}$ respectively when magnitude-scaled by 800 and frequency-scaled by 1000.

Step-by-step solution

Step 1 of 3

If the circuit is scaled by magnitude and frequency at the same time, the value of Resistance R' is,

$$R' = K_m R$$

Here, K_f is frequency scaling factor and R is the Resistance before scaling.

Substitute 800 for K_m and $12 \text{ k}\Omega$ for R .

$$\begin{aligned} R' &= (800)(12 \times 10^3) \text{ }\Omega \\ &= 96 \times 10^5 \text{ }\Omega \\ &= (96 \times 10^5 \text{ }\Omega) \left(\frac{1 \text{ M}\Omega}{1 \times 10^6 \text{ }\Omega} \right) \\ &= 9.6 \text{ M}\Omega \end{aligned}$$

Therefore, the value of R' is $\boxed{9.6 \text{ M}\Omega}$.

Step 2 of 3

If the circuit is scaled by magnitude and frequency at the same time, the value of Impedance L' is expressed as follows:

$$L' = \frac{K_m}{K_f} L$$

Here, K_m is magnitude scaling factor and L is the Inductance before scaling.

Substitute 800 for K_m , 1000 for K_f , and $40 \text{ }\mu\text{H}$ for L .

$$\begin{aligned} L' &= \left(\frac{800}{1000} \right) (40 \times 10^{-6}) \text{ H} \\ &= 32 \times 10^{-6} \text{ H} \\ &= (32 \times 10^{-6} \text{ H}) \left(\frac{1 \text{ }\mu\text{H}}{1 \times 10^{-6} \text{ H}} \right) \\ &= 32 \text{ }\mu\text{H} \end{aligned}$$

Therefore, the value of L' is $\boxed{32 \text{ }\mu\text{H}}$.

Step 3 of 3

If the circuit is scaled by magnitude and frequency at the same time, the value of Capacitance C' is,

$$C' = \frac{C}{K_m K_f}$$

Here, K_m is magnitude scaling factor, K_f is frequency scaling factor and C is the Capacitance before scaling.

Substitute 800 for K_m , 1000 for K_f , and 300 nF for C .

$$\begin{aligned} C' &= \frac{1}{(800)(1000)} (300 \times 10^{-9}) \text{ F} \\ &= 0.375 \times 10^{-12} \text{ F} \\ &= (0.375 \times 10^{-12} \text{ F}) \left(\frac{1 \text{ pF}}{1 \times 10^{-12} \text{ F}} \right) \\ &= 0.375 \text{ pF} \end{aligned}$$

Therefore, the value of C' is 0.375 pF.